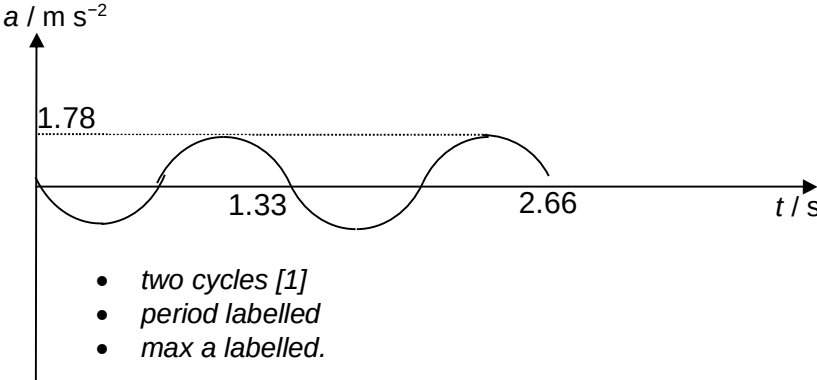


4	(a)	(i)	$\omega = \frac{45(2\pi)}{60}$ $= 4.71 \text{ rad s}^{-1}$	1 1
		(ii)	$a = r\omega^2 = (0.08)(4.712389)^2$ $= 1.78 \text{ m s}^{-2}$ towards the centre	1 1 1
	(b)	(i)	1. Period = $2\pi/T = 1.33 \text{ s}$	1
			2. Differentiation twice from $x = A \sin(\omega t)$ OR $a = -\omega^2 x = -\omega^2 A \sin(\omega t)$ input values : $a = -0.080 \times 22.2 \sin(4.71t) = -1.7765 \sin(4.71t)$ $= -1.78 \sin(4.71t)$	1 1
		(ii)	$a = -1.78 \sin(4.71 \times 0.20) = 1.44 \text{ m s}^{-2}$	1 1
		(iii)	$a / \text{m s}^{-2}$  • two cycles [1] • period labelled • max a labelled.	1 1 1
		(iv)	$0.060 \cos 4.71t$ Cosine expression [1] allow $\sin(4.71t - \pi/2)$ Correct values [1]	1 1

E	A	C		A
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